LAB 4

BACKPROP BY HAND

Understanding Gradient Descent - one gradient at a time



Harvard Computers, c. 1890. Rare Historical Photos

PRESS → OR SPACE. →

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Today's lab

We are the Computer

1 · Forward pass and loss together

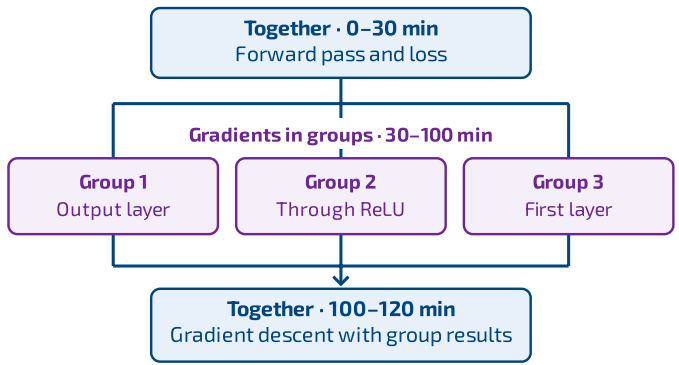
```
p = model(x)
loss = F.binary_cross_entropy(p, y)
```

2 · Calculate gradients by hand in groups

```
loss.backward()
```

3 · Update the parameters together

```
optimizer.step()
```

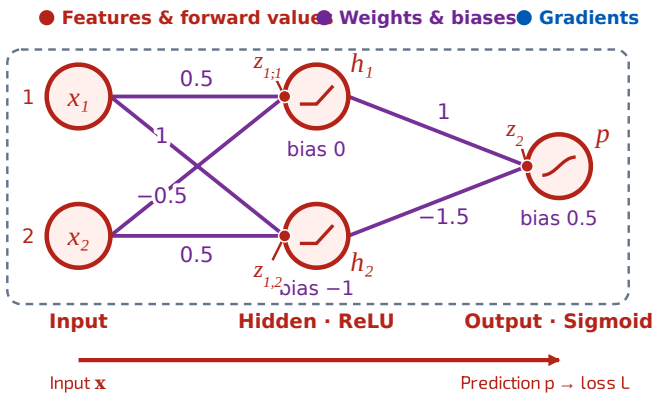


Repeated updates reduce the loss and let hidden layers learn useful features from data. This is central to representation learning.

David Rumelhart, Geoffrey Hinton, and Ronald Williams. *Learning representations by back-propagating errors*. *Nature* (1986).

Phase 1 · Forward pass

$$L = -\ln\Big(\underbrace{\sigma\big(\mathbf{w}_2^\top \underbrace{\mathrm{ReLU}(\overbrace{\mathbf{W}_1\mathbf{x} + \mathbf{b}_1}^{\mathbf{z}_1})}_{\mathbf{h}} + b_2\big)}_{p}\Big)$$



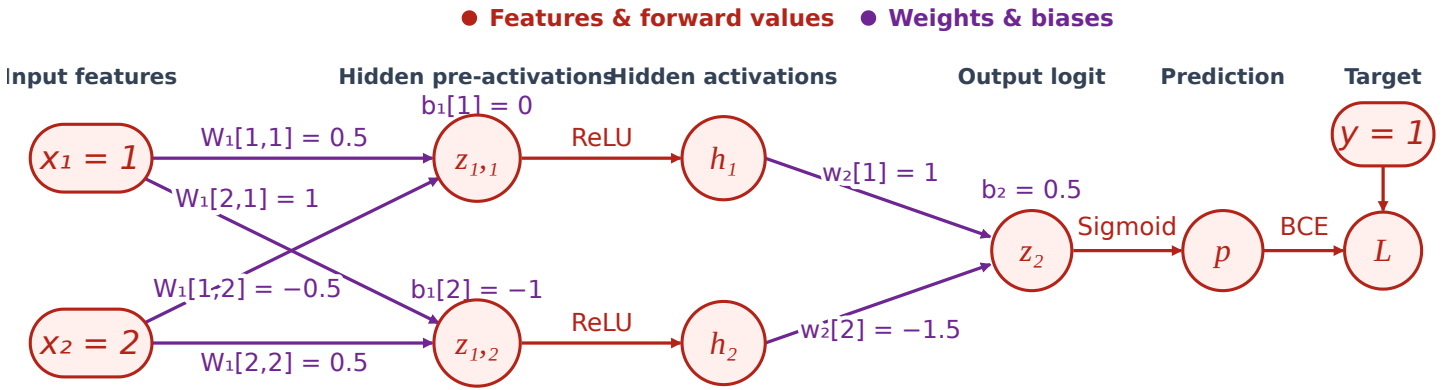
Use $\mathbf{x} = (1, 2)^\top$, $y = 1$, and the parameters in the diagram.

1. **Hidden layer.** Compute $\mathbf{z}_1 = \mathbf{W}_1\mathbf{x} + \mathbf{b}_1$ and $\mathbf{h} = \mathrm{ReLU}(\mathbf{z}_1)$.
2. **Output layer.** Compute $z_2 = \mathbf{w}_2^\top \mathbf{h} + b_2$ and $p = \sigma(z_2)$.
3. **Loss.** Compute $L = -\ln p$ for $y = 1$.

Record $\mathbf{z}_1$, $\mathbf{h}$, z_2, p, and L to three decimal places for the group calculations.

Phase 1: Let's calculate the Forward Pass together

Use $\mathbf{x} = (1, 2)^\top$ and $y = 1$. Round to three decimal places.

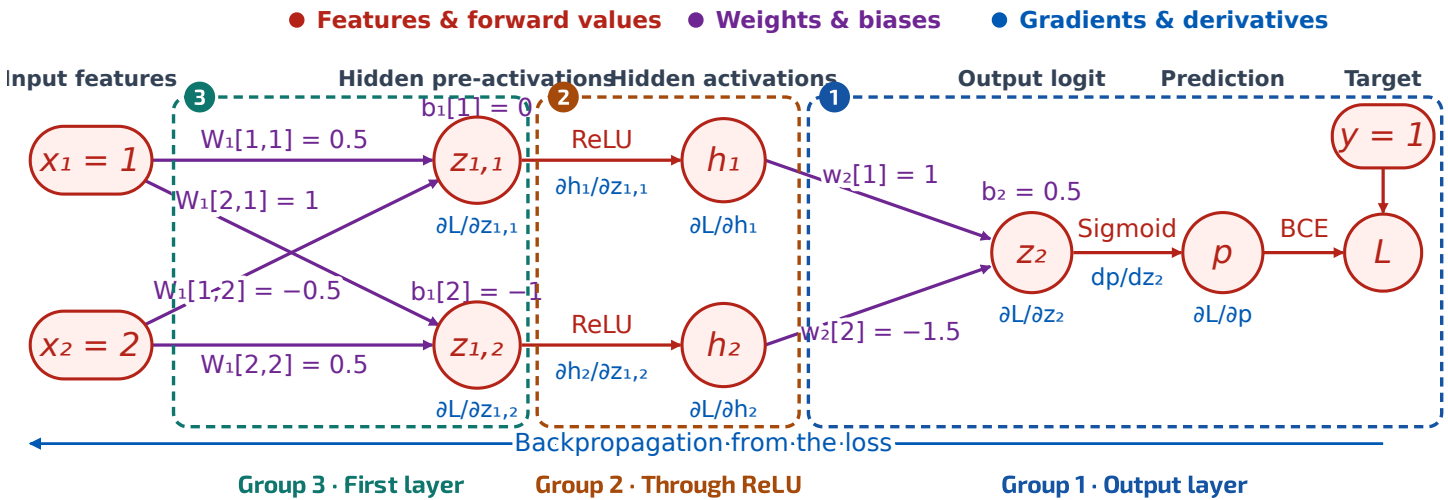


Click a node to enter a value · click again to toggle symbol / number · double-click to edit

Show numbers Reset

Phase 2 · Calculate gradients in three groups

Open your group's slide and use its starting values. All three groups can begin at the same time.

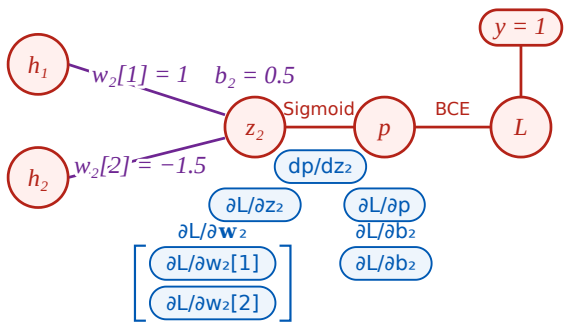


Bring your derivation and results to the whole-class discussion.

Group 1 · Output layer

$$L = \mathrm{BCE}\Big(\underbrace{\sigma\big(\overbrace{\mathbf{w}_2^\top \mathbf{h} + b_2}^{z_2}\big)}_{p}, \, y\Big)$$

From the loss L to $\partial L / \partial z_2$ and the gradients for $\mathbf{w}_2$ and b_2.



Start with $\mathbf{h} = (0, 1)^\top$, $p \approx 0.269$, and $y = 1$.

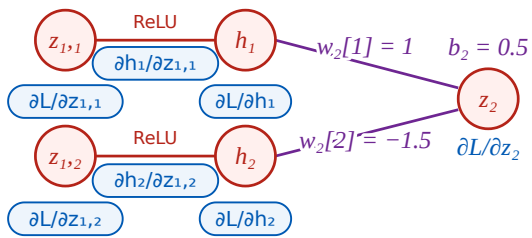
1. Find $\partial L / \partial p$ for binary cross-entropy and dp/dz_2 for the sigmoid.
2. Multiply and simplify for general binary y to get $\partial L / \partial z_2$. Then substitute the starting values.
3. Differentiate $z_2 = \mathbf{w}_2^\top \mathbf{h} + b_2$ with respect to each output weight and the output bias. Apply the chain rule.

Expected outputs · $\partial L / \partial p$, dp/dz_2, $\partial L / \partial z_2$, both entries of $\partial L / \partial \mathbf{w}_2$, and $\partial L / \partial b_2$. Click a blue label or matrix entry, then press **Enter** to check. Double-click to edit.

Group 2 · Through ReLU

$$z_2 = \mathbf{w}_2^\top \underbrace{\mathrm{ReLU}(\mathbf{z}_1)}_{\mathbf{h}} + b_2$$

From $\partial L/\partial z_2$ back through the hidden activations to $\partial L/\partial \mathbf{z}_1$.



Start with $\partial L/\partial z_2 \approx -0.731$, $\mathbf{z}_1 = (-0.5, 1)^\top$, $\mathbf{w}_2 = (1, -1.5)^\top$, and $b_2 = 0.5$.

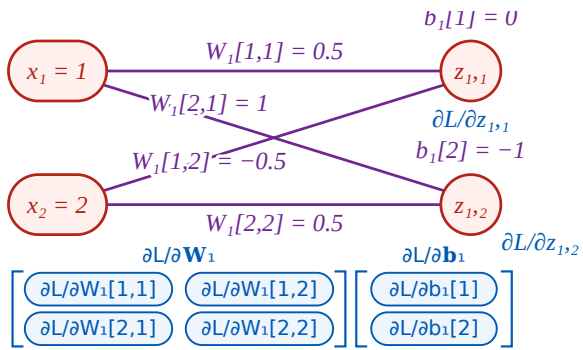
1. Differentiate $z_2 = \mathbf{w}_2^\top \mathbf{h} + b_2$ with respect to each entry of $\mathbf{h}$. Use $\partial L/\partial z_2$ to get $\partial L/\partial \mathbf{h}$.
2. Find the two ReLU derivatives, $\partial h_1/\partial z_{1,1}$ and $\partial h_2/\partial z_{1,2}$, at the given pre-activations.
3. Multiply each unit's loss gradient by its ReLU derivative to get $\partial L/\partial \mathbf{z}_1$.

Expected outputs · Both ReLU slopes and both entries of each gradient, $\partial L/\partial \mathbf{h}$ and $\partial L/\partial \mathbf{z}_1$. Click a blue label or matrix entry, then press **Enter** to check. Double-click to edit.

Group 3 · First layer

$$\underbrace{\mathbf{W}_1\mathbf{x} + \mathbf{b}_1}_{\mathbf{z}_1} = (z_{1,1}, z_{1,2})^\top$$

From $\partial L/\partial \mathbf{z}_1$ to the gradients for the first-layer weights and biases.

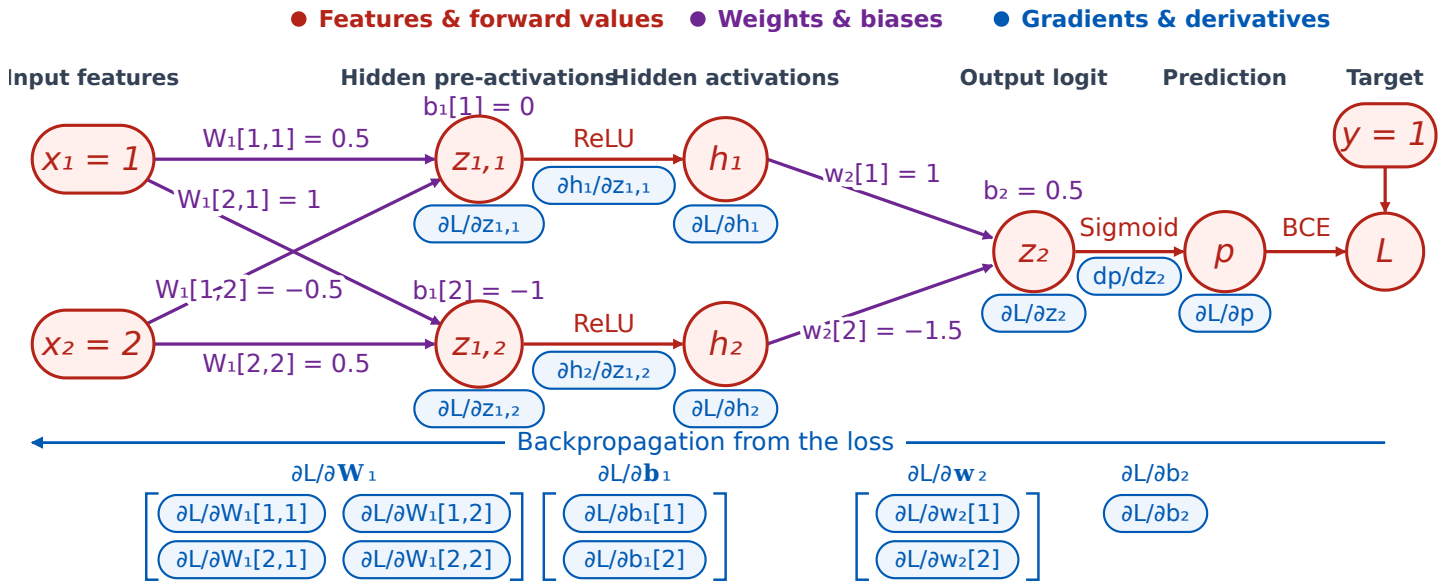


Start with $\partial L/\partial \mathbf{z}_1 \approx (0, 1.097)^\top$ and $\mathbf{x} = (1, 2)^\top$.

1. For hidden unit j, write $z_{1,j}$ in terms of row j of $\mathbf{W}_1$ and its bias.
2. Use the local derivatives and $\partial L/\partial z_{1,j}$ to find both weight gradients and the bias gradient for that unit. Repeat for the other unit.

Expected outputs · All four entries of $\partial L/\partial \mathbf{W}_1$ and both entries of $\partial L/\partial \mathbf{b}_1$. Click a blue label or matrix entry, then press **Enter** to check. Double-click to edit.

Phase 2 · Assemble the gradients



Click a blue gradient to enter a value · click again to toggle symbol / number · double-click to edit

Show numbers | Reset

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Phase 3 · Update and recompute together

$$\theta_{\text{new}} = \theta - 0.5\,\frac{\partial L}{\partial \theta}$$

Each group updates its three parameters. Fill any missing gradients, then enter each new value and press **Enter** to check it.

Group 1 · Output layer

Original	Loss gradient		New value
$w_2[1] = 1$	$\partial L/\partial w_2[1]$		
$w_2[2] = -1.5$	$\partial L/\partial w_2[2]$		
$b_2 = 0.5$	$\partial L/\partial b_2$		

Group 2 · Hidden unit 1

Original	Loss gradient		New value
$W_1[1,1] = 0.5$	$\partial L/\partial W_1[1,1]$		
$W_1[1,2] = -0.5$	$\partial L/\partial W_1[1,2]$		
$b_1[1] = 0$	$\partial L/\partial b_1[1]$		

Group 3 · Hidden unit 2

Original	Loss gradient		New value
$W_1[2,1] = 1$	$\partial L/\partial W_1[2,1]$		
$W_1[2,2] = 0.5$	$\partial L/\partial W_1[2,2]$		
$b_1[2] = -1$	$\partial L/\partial b_1[2]$		

Combine the nine updated parameters and run a fresh forward pass together. Compare the new loss and ReLU states with the original ones.